

A randomized fixed-parameter algorithm for the minimum quartet inconsistency problem *

Maw-Shang Chang[†], Ling-Ju Hung[‡], Chuang-Chieh Lin[§]

Department of Computer Science and Information Engineering,
National Chung Cheng University, Ming-Hsiung, Chiayi, Taiwan, R.O.C.

and

Peter Rossmanith[¶]

Department of Computer Science, RWTH Aachen University, Germany

Abstract

Given a set of n taxa S , exactly one topology for every subset of four taxa, and a positive integer k as the parameter, the parameterized Minimum Quartet Inconsistency (MQI) problem is to decide whether we can find an evolutionary tree inducing a set of quartet topologies that differs from the given set in at most k quartet topologies. The best fixed-parameter algorithm devised so far for the parameterized MQI problem runs in time $O(4^k n + n^4)$. In this paper, we present a randomized fixed-parameter algorithm that runs in time $O(3.47^k k^{3/2} n + n^4)$ with arbitrarily small constant error probability.

1 Introduction

Nowadays, to determine the evolutionary relationship of a set of taxa is very important in biological research, especially in computational biology. For this relationship, an *evolutionary tree* is a common model, which is widely considered. Let S be a set of taxa and $|S| = n$. An evolutionary tree T on S is an *unrooted*, *leaf-labeled* tree such that the leaves of T are bijectively labeled by the taxa in S , and each internal node of T has degree 3. In the past decade, *quartet methods* for building an evolutionary tree for a set of taxa have received much attention [5, 8].

*This work was carried out while Peter Rossmanith was visiting Department of Computer Science and Information Engineering, National Chung Cheng University in Autumn 2006.

[†]E-Mail: mschang@cs.ccu.edu.tw

[‡]E-Mail: hunglc@cs.ccu.edu.tw

[§]Corresponding author (E-Mail: lincc@cs.ccu.edu.tw)

[¶]E-Mail: rossmani@informatik.rwth-aachen.de

A *quartet* is a set of 4 taxa $\{a, b, c, d\}$ in the taxon set S . The *quartet topology* for $\{a, b, c, d\}$ induced by T is the path structure connecting a , b , c , and d in T (see Figure 1 for an illustration). A quartet $\{a, b, c, d\}$ has three possible topologies $[ab|cd]$, $[ac|bd]$ and $[ad|bc]$, which are the bipartitions of $\{a, b, c, d\}$ (hence $[ab|cd]$, $[ba|cd]$, $[ab|dc]$, $[ba|dc]$, $[cd|ab]$, $[dc|ab]$, $[cd|ba]$, $[dc|ba]$ are viewed as the same topologies of the quartet $\{a, b, c, d\}$). Figure 2 is an illustration for the three possible topologies of the quartet $\{a, b, c, d\}$.

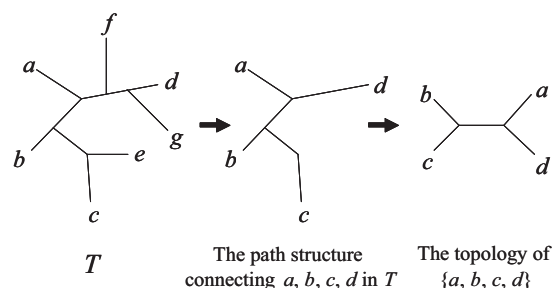


Figure 1: The quartet topology induced by an evolutionary tree T by $\{a, b, c, d\}$.

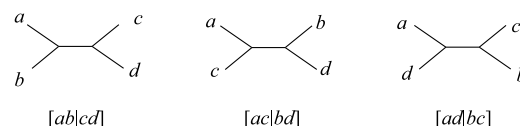


Figure 2: Three quartet topologies for the quartet $\{a, b, c, d\}$.

Suppose we are given a set S of n taxa and a set Q of quartet topologies. Let Q_T be the set of quar-

tet topologies induced by T . If there exists an evolutionary tree T such that $Q \subseteq Q_T$, we say that Q is *tree-consistent* (with T) or T *satisfies* Q . If there exists a tree T such that $Q = Q_T$, we say that Q is *tree-like* or *compatible*. Q is called *complete* if Q contains exactly one topology for every quartet, otherwise, we say Q is *incomplete*. For two complete sets of quartet topologies Q and Q^* where Q^* is tree-like but Q is not, the *quartet errors* of Q with respect to Q^* are the quartet topologies in Q that differ from those in Q^* . We denote the number of quartet errors of Q with respect to Q^* by $\Delta(Q, Q^*)$. Obviously, $0 \leq \Delta(Q, Q^*) \leq \binom{n}{4}$. In addition, we define the *number of quartet errors* of Q to be $\min\{\Delta(Q, Q^*) : Q^* \text{ is tree-like}\}$.

1.1 Related work

The optimization problem, called the *Maximum Quartet Consistency Problem* (MQC), is to construct an evolutionary tree T on S to satisfy as many quartet topologies of Q as possible. The *Minimum Quartet Inconsistency Problem* (MQI), which is a dual problem to the MQC problem, is to construct an evolutionary tree T on S such that the number of quartet errors of Q with respect to Q_T is minimized. The MQC problem is **NP**-hard [4], yet it has a *polynomial time approximation scheme* (PTAS) [9]. The MQI problem is also **NP**-hard [4], yet it admits an approximation algorithm with best ratio $O(n^2)$ [8] so far. If the input set of quartet topologies Q is not necessarily complete, Ben-Dor et al. gave an $O(3^nm)$ algorithm to solve the MQI problem by dynamic programming [2], where m is the number of quartet topologies.

Provided with a positive integer k as an additional part of the input, the parameterized MQI problem is to determine whether there exists an evolutionary tree T such that the number of quartet errors of Q is at most k . Gramm and Niedermeier proved that the parameterized MQI problem is *fixed parameter tractable* [7], that is, it can be solved in time $f(k)n^{O(1)}$ where f may be an arbitrary function only depending on k but not depending on n . They also gave a fixed parameter algorithm for the MQI problem with running time $O(4^kn + n^4)$ [7], where k is the number of quartet errors of Q . In [13], Wu et al. presented a lookahead branch-and-bound algorithm for solving the MQC problem with time complexity $O(4^{k'}n + n^4)$, where k' is an upper bound on the number of quartet errors of Q .

Besides, the *Quartet Compatibility Problem* (QCP) is to determine if there exists an evolution-

ary tree T on S satisfying all quartet topologies Q . The QCP problem can be solved in polynomial time if Q is complete [6]. But it becomes **NP**-complete if Q is incomplete [10].

1.2 Our result

In this paper, we focus on the parameterized MQI problem as follows. Given a complete set of quartet topologies Q and a parameter k as the input, determine whether there is a tree-consistent quartet set that differs from Q in at most k quartet topologies, that is, determine whether Q has at most k quartet errors. We propose a $O(3.47^k k^{3/2}n + n^4)$ randomized fixed-parameter algorithm with arbitrarily small constant error probability.

The paper is organized as follows. In Section 2, we give additional definitions, notations, required lemmas and theorems as preliminaries. In Section 3, a randomized fixed-parameter algorithm for the parameterized MQI problem will be introduced together with the analysis.

2 Preliminaries

Recall that S is a set of taxa and $|S| = n$. Let Q denote the complete set of quartet topologies over S . The set Q is of size $\binom{n}{4}$. Gramm and Niedermeier [7] showed that a complete set of quartet topologies is tree-consistent if and only if it is tree-like. If a set of quartet topologies is not tree-consistent, we say that it has a *conflict*. When a complete set of quartet topologies Q is not tree-consistent, we say it has a *global conflict*. In contrary, we call a size three set of topologies has a *local conflict* if it is not tree-consistent. Bandelt and Dress [3] introduced the so-called *substitution property* to identify the subsets of the quartet topologies which are not tree-consistent. The substitution property is introduced as follows.

Definition 1. [3] Given a set of taxa S and a complete set of quartet topologies Q over S , a set $S' \subseteq S$ of five taxa satisfies the *substitution property* if, for every choice of distinct 5 taxa $a, b, c, d, e \in S'$, $[ab|cd] \in Q$ implies $[ab|ce] \in Q$ or $[ae|cd] \in Q$.

By the substitution property, Bandelt and Dress gave the following proposition.

Proposition 1. [3] Given a set of taxa S and a complete set of quartet topologies Q and some

taxon $f \in S$, then Q is tree-like if and only if every set of size five that contains f satisfies the substitution property.

Gramm and Niedermeier [7] proved the following two useful lemmas concerning the tree-consistency of a set of three quartet topologies.

Lemma 1. [7] *Three quartet topologies involving more than five taxa are tree-consistent.*

Lemma 2. [7] *Three quartet topologies involving taxa from the taxon set S are tree-consistent if and only if they do not contradict the substitution property.*

By Lemma 1 and 2, we obtain that a set of three quartet topologies has a local conflict if it involves exactly five taxa and contradicts the substitution property. In order to check whether Q is tree-like, Gramm and Niedermeier showed that we can examine if every set of three quartet topologies from Q which involves some taxon f is tree-like. The result is listed as the following theorem.

Theorem 1. [7] *Given a set of taxa S and a complete set of quartet topologies Q over S , and some taxon $f \in S$, Q is tree-like if and only if every set of three quartet topologies from Q that involves f is tree-consistent.*

Gramm and Niedermeier obtained the observation as follows. For some taxon f in S , a complete set Q of quartet topologies is not tree-like if and only if Q has at least one local conflict involving f . Besides, all local conflicts involving f can be found in time $O(n^4)$. We summarize the above results in the following theorem.

Theorem 2. [7] *If we are given a set S of taxa, some taxon $f \in S$, and a complete set Q of quartet topologies that is not tree-consistent, then Q has at least one local conflict involving f and all local conflicts involving f can be found in time $O(n^4)$.*

For a quartet topology $[ab|cd]$, we call $[ab|cd]$ and $[cd|ab]$ its two *orientations*. Given a set of quartet topologies, where each of the topologies is assigned an orientation, let l be the number of different taxa occurring in the left-hand sides of the topologies and let r be the number of different taxa occurring in the right-hand sides of the topologies.

Definition 2. [7] The *signature* of the set of quartet topologies is the pair (l, r) that, over all possible orientations for these topologies, minimizes l .

For example, let $Q_1 = \{[ab|cd], [ab|de]\}$ and $Q_2 = \{[ab|cd], [be|cd], [ad|ce]\}$. The signature of Q_1 is $(2, 3)$ and the signature of Q_2 is $(3, 4)$. Through the signatures of the set of quartet topologies, Gramm and Niedermeier [7] proposed the following useful theorem to check if a set of three quartet topologies has a local conflict:

Theorem 3. [7] *Three quartet topologies are not tree-consistent iff they involve five taxa and their signature is $(3, 4)$ or $(4, 4)$.*

For the analysis of the following sections, we list the following well-known theorems here for convenience.

Theorem 4 (Stirling's Formula).

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + O\left(\frac{1}{n}\right)\right).$$

Theorem 5. *For any positive integer n and any positive constant $0 \leq \alpha \leq 1$,*

$$\binom{n}{\alpha n} = \frac{(1 + O(n^{-1}))}{\sqrt{2\pi\alpha(1-\alpha)n}} \left(\frac{1}{\alpha^\alpha(1-\alpha)^{1-\alpha}}\right)^n.$$

Proof. By Stirling's formula (see Theorem 4), we have

$$\begin{aligned} \binom{n}{\alpha n} &= \frac{n!}{(\alpha n)!((1-\alpha)n)!} \\ &= \sqrt{2\pi n} \left(\frac{n}{e}\right)^n (1 + O(n^{-1})) \\ &\quad \cdot \frac{1}{\sqrt{2\pi\alpha n}(\alpha n/e)^{\alpha n}} \\ &\quad \cdot \frac{1}{\sqrt{2\pi(1-\alpha)n}((1-\alpha)n/e)^{(1-\alpha)n}} \\ &\quad \cdot \frac{1}{(1 + O(n^{-1}))^2} \\ &= \frac{(1 + O(n^{-1}))}{\sqrt{2\pi\alpha(1-\alpha)n}} \left(\frac{1}{\alpha^\alpha(1-\alpha)^{1-\alpha}}\right)^n. \end{aligned}$$

□

3 A randomized fixed-parameter algorithm

3.1 The algorithm

We use the RAM model of computation. For each $t \in Q$, we can access t in one time unit. Here

we use t' and t'' to denote the two alternative quartet topologies of each quartet topology $t \in Q$. We devise a randomized fixed-parameter algorithm for the parameterized MQI problem. The algorithm is composed of two main parts: Algorithm RandBranch and Algorithm RandFPT-MQI.

The concepts of the algorithm are as follows. First, we build a list of local conflicts C involving some arbitrarily chosen taxon f in Q . For each local conflict $c = (t_1, t_2, t_3) \in C$ which contradicts the substitution property that $t_1 \in Q \Rightarrow t'_2 \in Q$ or $t'_3 \in Q$, we record the alternative topologies t'_2 and t'_3 of t_2 and t_3 respectively for c in the local conflict list so that we can fix c (i.e., to eliminate the local conflict c) by changing t_2 to t'_2 , changing t_3 to t'_3 or changing t_1 to its two alternative topologies t'_1 and t''_1 . Second, we repeatedly choose a local conflict c in C and then try to fix it until the local conflict list C gets to empty.

Algorithm 1 : RandFPT-MQI.

```

1: Build a list of local conflicts  $C$  involving some
   arbitrary chosen taxon  $f$  in  $Q$ .
2: repeat
3:   if RandBranch( $Q, k, C$ ) = TRUE then
4:     Return PASS;
5:   end if
6: until the loop has been executed  $m$  times
7: Return REJECT.
```

3.2 Analysis of the algorithm

3.2.1 Correctness

By Theorem 1, the remaining set of quartet topologies Q' produced by Algorithm RandFPT-MQI is tree-like iff there no local conflict remained in Q' . Besides, Algorithm RandFPT-MQI returns ACCEPT (i.e., once Algorithm RandBranch returns TRUE) only if all local conflicts are resolved and if Q' differs from Q in at most k quartet topologies. Therefore, if Algorithm returns ACCEPT, then the input Q must have no more than k quartet errors with respect to some tree-like set of quartet topologies over S (i.e., $\Delta(Q, Q^*) \leq k$ for some tree-like set of quartet topologies Q^*). In other words, if there is no tree-like set Q^* of quartet topologies such that $\Delta(Q, Q^*) \leq k$, Algorithm RandFPT-MQI must return REJECT. Yet if Algorithm RandFPT-MQI returns REJECT, it might err with small probability. That is, Algorithm RandFPT-MQI might fail with small probability to return ACCEPT for the input Q which has no more

Algorithm 2 : RandBranch (a complete set of quartet topologies Q , an integer parameter k and a local conflict list C).

```

1:  $Q' \leftarrow Q$ ;
2: repeat
3:   if  $C = \emptyset$  and  $Q'$  differs from  $Q$  in at most  $k$ 
     topologies then
4:     Return TRUE.
5:   else if  $C = \emptyset$  then
6:     Return FALSE.
7:   else
8:     Arbitrarily choose a local conflict  $c =$ 
        $\{t_1, t_2, t_3\} \in C$  such that  $c$  violates the
       substitution property that " $t_1 \in Q' \Rightarrow$ 
       ( $t'_2 \in Q'$  or  $t'_3 \in Q'$ )".
9:     Choose one of the following 4 subcases
       uniformly at random and execute it:

       (i)    $C \leftarrow \text{update}(C, t'_1);$ 
              $Q' \leftarrow Q' - t_1 + t'_1;$ 

       (ii)   $C \leftarrow \text{update}(C, t''_1);$ 
              $Q' \leftarrow Q' - t_1 + t''_1;$ 

       (iii)  $C \leftarrow \text{update}(C, t'_1);$ 
              $Q' \leftarrow Q' - t_2 + t'_2;$ 

       (iv)   $C \leftarrow \text{update}(C, t'_3);$ 
              $Q' \leftarrow Q' - t_3 + t'_3.$ 

10:  end if
11: until the loop has been executed  $2k$  times.
12: Return FALSE.
```

than k quartet errors. Thus Algorithm RandFPT-MQI is a Monte Carlo algorithm. Its running time and error probability will be analyzed in the following subsection.

3.2.2 Time complexity and error probability

Building the local conflict list: By Theorem 2, we can build a local conflict list involving some fixed taxa f in $O(n^4)$ time.

Testing emptiness of C and the number of quartet errors with respect to some tree-like Q' : The testings at line 3 in Algorithm RandBranch involve testing whether C is empty and testing if $\Delta(Q, Q') \leq k$. The former can be easily done in constant time. The latter can be done also in constant time by updating the number of quartet topologies of Q' that differ from those in Q in each iteration of the algorithm.

The procedure update: Let t be a quartet topology involving a quartet $\{a, b, c, d\}$. Whenever we change t to another topology t' , we have to update the local conflict list. This is done by the instruction $\text{update}(C, t')$. We only have to check the sets of five taxa involving a, b, c , and d . The number of these sets is $O(n)$. By Theorem 3, we have that checking whether a set of three quartet topologies is tree-consistent requires only constant time, hence $\text{update}(C, t')$ can be done in $O(n)$ time.

Probabilistic analysis of the algorithm:

Let Q^* be a fixed tree-like set of quartet topologies over S . Let us consider the following lemma.

Lemma 3. *Given two complete sets of quartet topologies Q and Q^* over the taxon set S , where Q^* is tree-like and Q is not. Let C be the local conflict list of Q and let $c = \{t_1, t_2, t_3\} \in C$ be an arbitrary local conflict in Q . Assume that $\Delta(Q, Q^*) = k$ and c does not satisfy the substitution property that “ $t_1 \in Q$ implies $t'_2 \in Q$ or $t'_3 \in Q$ ”, then for $Q_1 = Q - t_1 + t'_1$, $Q_2 = Q - t_1 + t''_1$, $Q_3 = Q - t_2 + t'_2$ and $Q_4 = Q - t_3 + t'_3$, we have*

- (i) $\Delta(Q_i, Q^*) \leq k + 1$ for each $i \in \{1, 2, 3, 4\}$ and
- (ii) $\Delta(Q_i, Q^*) = k - 1$ for some $i \in \{1, 2, 3, 4\}$.

Proof. (i) Since Q_i is obtained by changing exactly one quartet topology in Q for each $i \in \{1, 2, 3, 4\}$, the number of quartet errors in Q_i will be increased by at most 1. Hence this part of the lemma follows.

- (ii) Since $c = \{t_1, t_2, t_3\}$ contradicts to the substitution property that $t_1 \in Q$ implies $t'_2 \in Q$ or $t'_3 \in Q$, we have $t_1 \in Q$, $t'_2 \notin Q$ and $t'_3 \notin Q$. Consider the following two cases.

- (a) Assume that $t_1 \in Q^*$. Since Q^* is tree-like, we have $t'_2 \in Q^*$ or $t'_3 \in Q^*$ by the substitution property. If $t'_2 \in Q^*$, then $t_2 \notin Q^*$ since Q^* is complete. Then we have $\Delta(Q_3, Q^*) = k - 1$ since $t_2 \in Q$ and $t_2 \notin Q^*$ but $t'_2 \in Q^*$. Otherwise, $t_3 \in Q^*$, and then similarly we have $\Delta(Q_4, Q^*) = k - 1$.
- (b) Otherwise, $t_1 \notin Q^*$. Then we have $t'_1 \in Q^*$ or $t''_1 \in Q^*$ since Q^* is complete. If $t'_1 \in Q^*$, then $t_1 \notin Q^*$. We have $\Delta(Q_1, Q^*) = k - 1$ since $t_1 \in Q$ and $t_1 \notin Q^*$ but $t'_1 \in Q^*$. Otherwise, $t''_1 \in Q^*$, so $t_1 \notin Q^*$ since Q^* is complete. Similarly we have $\Delta(Q_2, Q^*) = k - 1$.

From the results in the above cases, the second part of the lemma immediately follows. $\square$

Let Q'_t be Q' after t iterations of Algorithm RandBranch with input set of quartet topologies Q . Note that $Q'_0 = Q$. Let Q^* be a tree-like quartet topology set Q such that $\Delta(Q, Q^*)$ is minimum and let $^1 k = \Delta(Q, Q^*)$. We say Q^* is a *nearest tree-like quartet topology set* to Q . In order to determine the number of executions of Algorithm RandBranch (i.e., the value of m), a lower bound on the probability that Algorithm RandBranch returns ACCEPT is required. This can be obtained by analyzing the probability that Algorithm RandBranch gets to Q^* in $2k$ iterations. First, let us consider the following lemma.

Lemma 4. *Let Q^* be a nearest tree-like quartet topology set to Q and assume that $\Delta(Q, Q^*) = k$. The probability that $\Delta(Q'_k, Q^*) \leq k/2$ is at least $O(k^{-1/2}3^{-k/2})$.*

Proof. Inspired by the analysis in Chapter 7 of [11] and Schöning’s work for k -SAT problem [12], we have the following analysis.

Since Q^* is a set of quartet topologies such that $\Delta(Q, Q^*) = k$ is minimum, Algorithm RandBranch will never visit a tree-like quartet topology set except Q^* or another that differs from Q by exactly k quartet topologies, and hence will not stop within the first k iterations. By Lemma 3, $\Delta(Q'_{i+1}, Q^*) = \Delta(Q'_i, Q^*) - 1$ with probability at least $1/4$ and $\Delta(Q'_{i+1}, Q^*) \leq \Delta(Q'_i, Q^*) + 1$ with probability at most $3/4$ for each integer $i \geq 0$. In the i th iteration of Algorithm RandBranch, we denote the change from Q'_i to Q'_{i+1} as a *right change* if $\Delta(Q'_{i+1}, Q^*) = \Delta(Q'_i, Q^*) - 1$, and we denote it by a *wrong change* if $\Delta(Q'_{i+1}, Q^*) \geq \Delta(Q'_i, Q^*)$. Let $f(k) = \binom{k}{k/4} (1/4)^{3k/4} (3/4)^{k/4}$. It is obvious that $f(k)$ is the probability of exactly $k/4$ wrong changes and $3k/4$ right changes in the first k iterations of Algorithm RandBranch. After $k/4$ wrong changes and $3k/4$ right changes, we have $\Delta(Q'_k, Q^*) \leq k/2$, thus $f(k)$ is a lower bound on $\Delta(Q'_k, Q^*) \leq k/2$.

Since $f(k) = \binom{k}{k/4} (1/4)^{3k/4} (3/4)^{k/4}$, by Theorem 4 and 5 we have

¹Note that it is safe to let $k = \Delta(Q, Q^*)$ for our analysis of success probability of the algorithm. The reason is as follows. If $k < \Delta(Q, Q^*)$, the algorithm will never return ACCEPT, and if $k > \Delta(Q, Q^*)$, the algorithm will return ACCEPT with higher probability.

$$\begin{aligned}
f(k) &= \frac{(1/4)^{3k/4}(3/4)^{k/4}(1+O(1/k))}{((1/4)^{1/4}(3/4)^{3/4})^k \sqrt{2\pi(1/4)(3/4)k}} \\
&= \frac{4}{\sqrt{6\pi}} \frac{1}{3^{k/2}\sqrt{k}} (1+O(1/k))
\end{aligned}$$

Hence we have $f(k) = \Omega(k^{-1/2}3^{-k/2})$. Therefore, the probability that $\Delta(Q'_k, Q^*) \leq k/2$ is at least $\Omega(k^{-1/2}3^{-k/2})$. $\square$

Now, let us see the following theorem.

Theorem 6. *Let $\Delta(Q, Q^*) = k$, where Q^* is a nearest tree-like quartet topology set. Then the success probability of Algorithm RandBranch is at least $\Omega(k^{-1/2}3^{-k/2}4^{-k/2})$.*

Proof. Let $Q^\#$ denote the Q'_k such that $\Delta(Q'_k, Q^*) \leq k/2$. Under the condition that Algorithm RandBranch gets to $Q^\#$ right after the k th iteration, the probability that the algorithm gets to Q^* is at least the probability that $\Delta(Q^\#, Q^*)$ successive right changes are performed by the algorithm *after* it reached $Q^\#$, so we have

$$\begin{aligned}
&\Pr \left[\Delta(Q'_i, Q^*) = 0 \text{ for some } k < i \leq 2k \right] \\
&\text{Algorithm RandBranch reached } Q^\# \geq \left(\frac{1}{4} \right)^{\frac{k}{2}},
\end{aligned}$$

By Lemma 4, we have the probability of $\Delta(Q'_k, Q^*) \leq k/2$ is $\Omega(k^{-1/2}3^{-k/2})$. Hence the success probability of Algorithm RandBranch is at least $\Omega(k^{-1/2}3^{-k/2}) \cdot 4^{-k/2} = \Omega(k^{-1/2}3^{-k/2}4^{-k/2})$. $\square$

Since procedure `update` costs $O(n)$ time, line 3 of Algorithm RandBranch requires only constant time, and the number of repetitions of the loop in Algorithm RandBranch is $2k$, it can be derived that the running time of Algorithm RandBranch is $O(kn)$. Besides, building a local conflict list requires $O(n^4)$ time. Let $m = ck^{1/2}3^{k/2}4^{k/2}$ for some constant c . Therefore, the running time of Algorithm RandFPT-MQI is $O(3^{k/2}4^{k/2}k^{3/2}n + n^4) = O(3.47^k k^{3/2}n + n^4)$. Moreover, since the success probability of Algorithm RandBranch is at least $4 \cdot 4^{-k/2}3^{-k/2}(1+O(1/k))/\sqrt{6\pi k}$, the error probability of Algorithm RandFPT-MQI is at most

$$\begin{aligned}
&\left(1 - \frac{4 \cdot 4^{-k/2}3^{-k/2}}{\sqrt{6\pi k}} (1+O(1/k)) \right)^{ck^{1/2}3^{k/2}4^{k/2}} \\
&\leq e^{-4c(1+O(1/k))/\sqrt{6\pi}} \\
&= e^{-4c/\sqrt{6\pi}} \cdot e^{-O(1/k)} \\
&= e^{-4c/\sqrt{6\pi}} (1+O(1/k)).
\end{aligned}$$

Therefore we have the following concluding theorem.

Theorem 7. *Algorithm RandFPT-MQI runs in time $O(3.47^k k^{3/2}n + n^4)$ with arbitrarily small constant error probability.*

References

- [1] P. Buneman: The recovery of trees from measures of dissimilarity. in: Hodson, et al. (Eds.), *Anglo-Romanian Conference on Mathematics in the Archaeological and Historical Sciences*, Edinburgh University Press, Edinburgh, 1971, 387–395.
- [2] A. Ben-Dor, B. Chor, D. Graur, R. Ophir, and D. Pelleg: From four-taxon trees to phylogenies: the case of mammalian evolution. In *Proceedings of the RECOMB*, 1998, 9–19.
- [3] H.-J. Bandelt and A. Dress: Reconstructing the shape of a tree from observed dissimilarity data. *Adv. in Appl. Math.* **7** (1986) 309–343.
- [4] V. Berry, T. Jiang, P. E. Kearney, M. Li, and H. T. Wareham: Quartet cleaning: Improved algorithms and simulations. In *Proceedings of the 7th Annual European Symposium on Algorithms (ESA 99)*, Lecture Notes in Comput. Sci., Vol. 1643, Springer Berlin, 1999, 313–324.
- [5] B. Cho: From quartets to phylogenetic trees. In *Proceedings of the 25th Conference on Current Trends in Theory and Practice of Informatics (SOFSEM)*, Lecture Notes in Comput. Sci., Vol. 1521, Springer, Berlin, 1998, 36–53.
- [6] P. Erdős, M. Steel, L. Székely, and T. Warnow: A few logs suffice to build (almost) all trees (Part 1). *Random Struct. Alg.* **14** (1999) 153–184.
- [7] J. Gramm and R. Niedermeier: A fixed-parameter algorithm for minimum quartet inconsistency. *J. Comput. System Sci.* **67** (2003) 723–741.

- [8] T. Jiang, P. E. Kearney, and M. Li: Some open problems in computational molecular biology. *J. Algorithms* **34** (2000) 194–201.
- [9] T. Jiang, P. E. Kearney, and M. Li: A polynomial time approximation scheme for inferring evolutionary tree from quartet topologies and its application. *SIAM J. Comput.* **30** (2001) 1942–1961.
- [10] M. Steel: The complexity of reconstructing trees from qualitative characters and subtrees. *J. Classification* **9** (1992) 91–116.
- [11] M. Mitzenmacher and E. Upfal: *Probability and Computing: Randomized Algorithms and Probabilistic Analysis*. Cambridge University Press, 2005.
- [12] U. Schöning: A probabilistic algorithm for k -SAT and constraint satisfaction problem. In *Proceedings of the 40th IEEE Symposium on the Foundations of Computer Science*, 1999, 410–414; *Algorithmica* **32** (2002) 615–623.
- [13] G. Wu, J-H. You, and G. Lin: A lookahead branch-and-bound algorithm for the maximum quartet consistency problem. In *Proceedings of WABI 2005*, Lecture Notes in Comput. Sci., Vol. 3692, 2005, 65–76.